ENGINE REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

id0110q2800400

Warning

- **A hot engine can cause severe burns. Turn off the engine and wait until it is cool before servicing.**
- **Fuel vapor is hazardous. It can very easily ignite, causing serious injury and damage. Always keep sparks and flames away from fuel.**
- **Fuel line spills and leakage are dangerous. Fuel can ignite and cause serious injuries or death and damage. Fuel can also irritate skin and eyes. To prevent this, always complete the "Fuel Line Safety Procedure". (See BEFORE SERVICE PRECAUTION [SKYACTIV-D 1.5].)**

Caution

- **Secure the steering wheel using tape or a cable to prevent the steering shaft from rotating after disconnecting the steering shaft. If the steering wheel rotates after the steering shaft and the steering gear and linkage are disconnected, the internal parts of the clock spring could be damaged.**

Note

- Perform the engine and transaxle component removal/installation from the lower side of the vehicle after removing the front crossmember component.

Operation After Replacing Engine

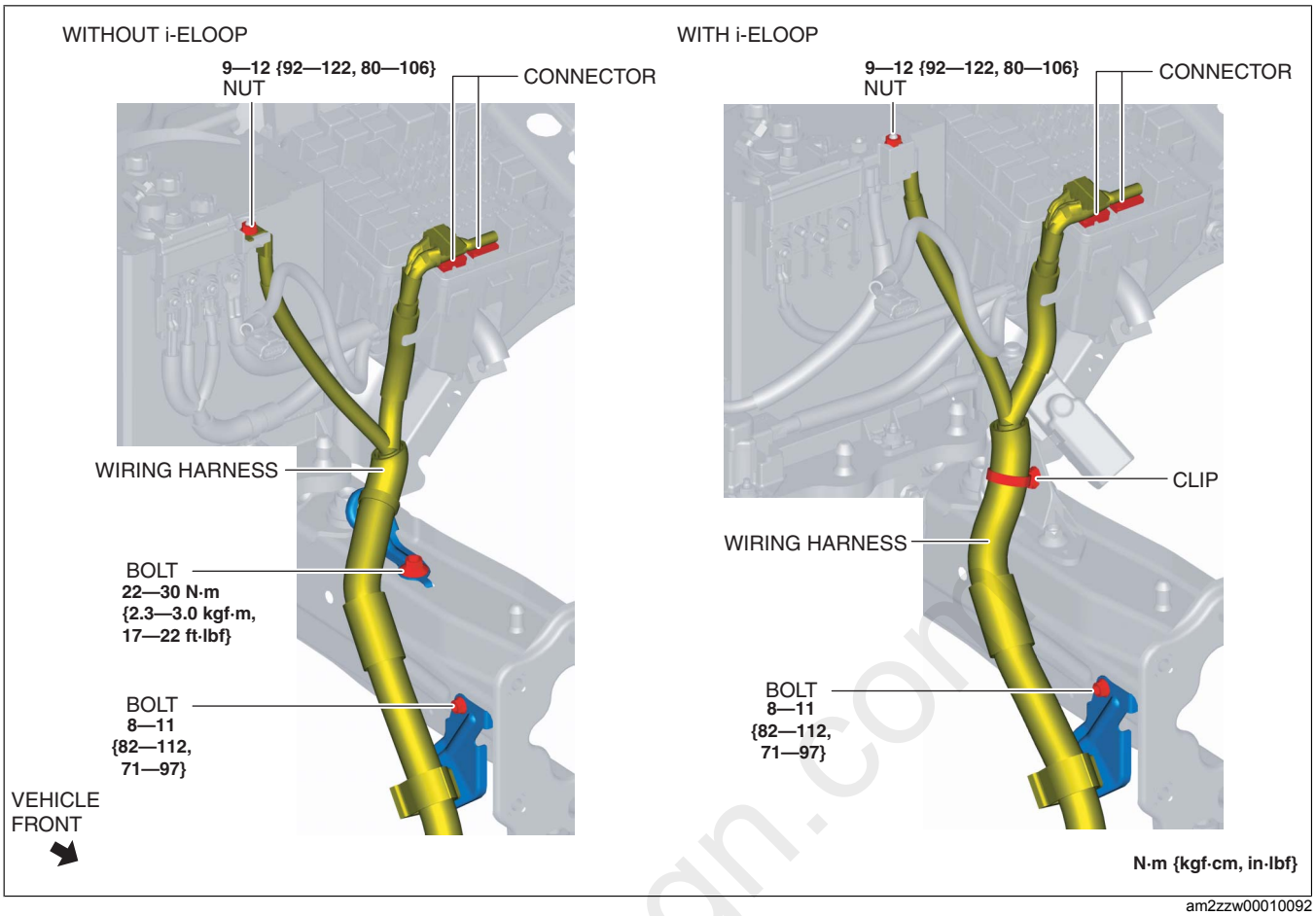
1. After replacing the engine, perform the following procedure.

STEP	ACTION	PAGE/CONDITION
1	Perform KOEO self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
2	Switch the ignition off.	—
3	Verify that the check engine light does not illuminate.	—
4	Perform KOER self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
5	Perform fuel injector injection amount correction.	(See FUEL INJECTOR INJECTION AMOUNT CORRECTION [SKYACTIV-D 1.5].)
6	Perform timing chain learning procedure.	(See TIMING CHAIN LEARNING [SKYACTIV-D 1.5].)
7	Clear the DTCs.	(See OBD READ/CLEAR DIAGNOSTIC TEST RESULTS [SKYACTIV-D 1.5].)
8	Switch the ignition off.	—

Removal/Installation

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the air cleaner and air hose as a single unit. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Remove the bolts, nut and connector shown in the figure, and set the wiring harness aside.

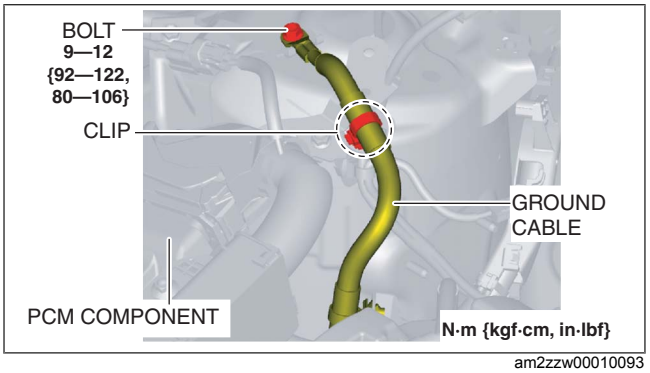
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5. Remove the battery and battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Remove the DC-DC converter (i-ELOOP) with the cables connected and secure the DC-DC converter (i-ELOOP) using wire or rope so that it is out of the way. (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
7. Disconnect the ground cable shown in the figure.

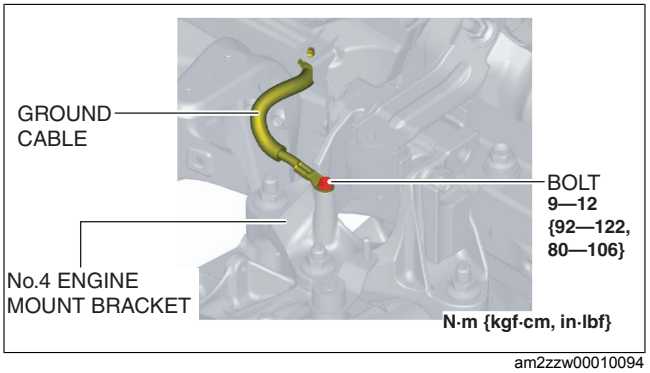
Right side



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Left side

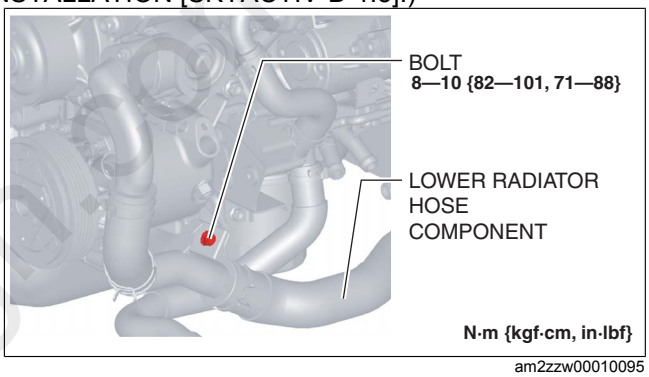
8. Remove the air inlet pipe. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
9. Drain the engine coolant and the coolant in the water-cooled charge air cooler. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].) (See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
10. Drain the transaxle oil (MTX) or ATF (ATX). (See MANUAL TRANSAXLE OIL REPLACEMENT [F66M-R].) (See AUTOMATIC TRANSAXLE FLUID (ATF) REPLACEMENT [EW6A-EL].)



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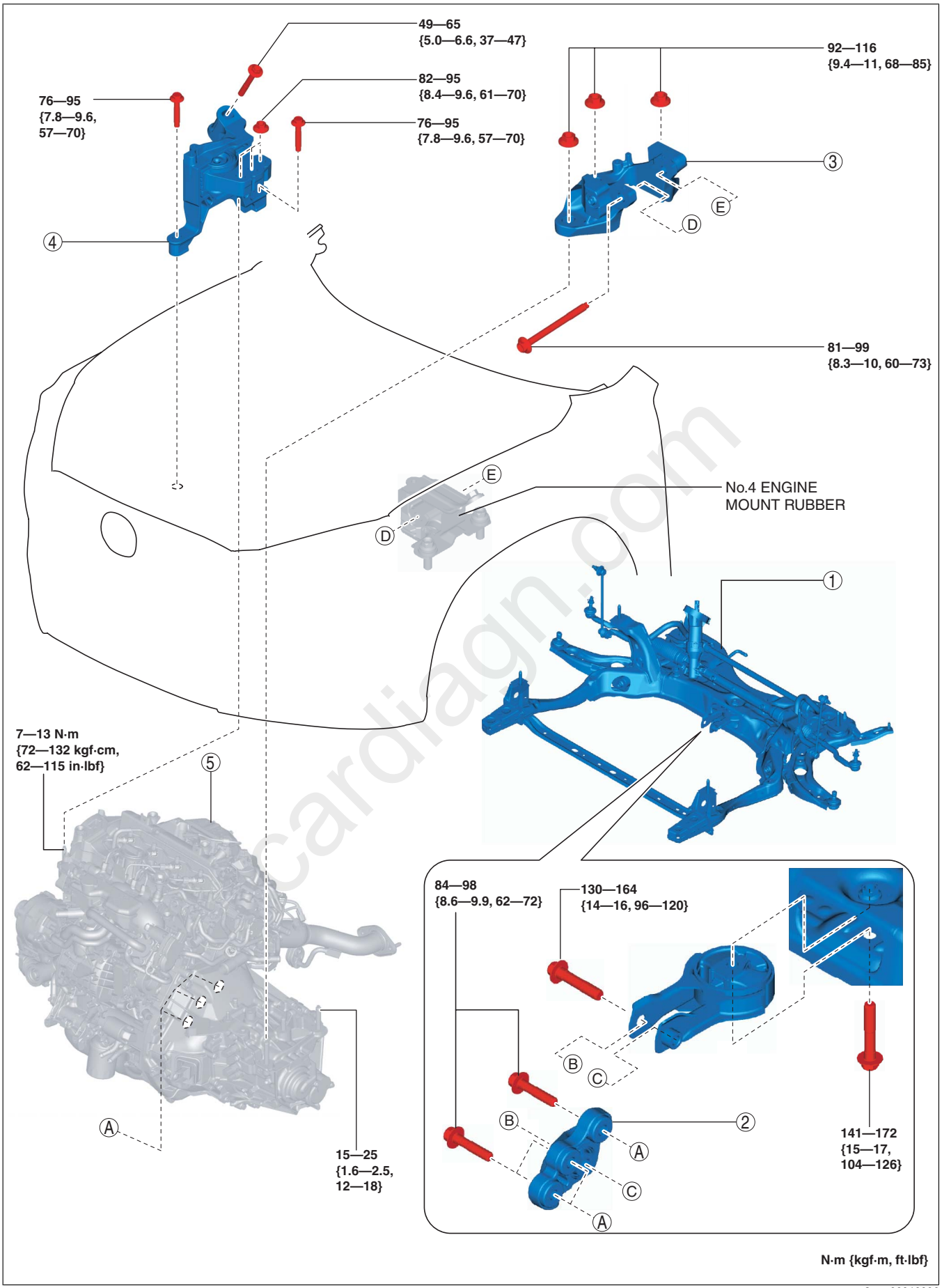
11. Remove the coolant reserve tank. (See COOLANT RESERVE TANK REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
12. Remove the cooling fan component. (See COOLING FAN MOTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
13. Disconnect the selector cable. (ATX) (See AUTOMATIC TRANSAXLE SHIFT MECHANISM REMOVAL/INSTALLATION.)
14. Disconnect the control cable. (MTX) (See MANUAL TRANSAXLE SHIFT MECHANISM REMOVAL/INSTALLATION [F66M-R].)
15. Remove the clutch release cylinder with the pipe still connected. (MTX) (See CLUTCH RELEASE CYLINDER REMOVAL/INSTALLATION [F66M-R].)
16. Disconnect the vacuum hose on the engine side. (See VACUUM HOSE REMOVAL/INSTALLATION [L.H.D.].) (See VACUUM HOSE REMOVAL/INSTALLATION [R.H.D.].)
17. Disconnect the fuel main hose and fuel return hose No.1 from lower case. (See LOWER CASE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
18. Disconnect the water-cooled charge air cooler. (upper, lower) (See WATER-COOLED CHARGE AIR COOLER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
19. Disconnect the heater hose from body side. (See A/C UNIT REMOVAL/INSTALLATION.)
20. Remove the front wheels and tires. (See GENERAL PROCEDURES (SUSPENSION).)
21. Remove the front under cover No.1 and No.2. (See FRONT UNDER COVER No.1 REMOVAL/INSTALLATION.) (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
22. Remove the front splash shield. (See SPLASH SHIELD REMOVAL/INSTALLATION.)
23. Remove the drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
24. Remove the bolt of the lower radiator hose component.
25. Remove the A/C compressor with the pipes connected and secure the A/C compressor using wire or rope so that it is out of the way. (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
26. Remove the middle pipe installation nut (catalytic converter side) and secure the middle pipe using wire or rope so that it is out of the way. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



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27. Disconnect the front drive shaft from the transaxle side and set it aside. (See FRONT DRIVE SHAFT REMOVAL/INSTALLATION.)
28. Remove in the order indicated in the table.
29. Install in the reverse order of removal.
30. Refill the transaxle oil (MTX) or ATF (ATX). (See MANUAL TRANSAXLE OIL REPLACEMENT [F66M-R].) (See AUTOMATIC TRANSAXLE FLUID (ATF) REPLACEMENT [EW6A-EL].)
31. Refill with engine coolant and water-cooled charge air cooler coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].) (See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
32. If the engine is replaced, perform “Operation After Replacing Engine”. (See Operation After Replacing Engine.)
33. Start the engine, and inspect and adjust the following:
    - Front wheel alignment (See FRONT WHEEL ALIGNMENT.)
    - Leakage of engine oil, engine coolant, water-cooled charge air cooler coolant, transaxle oil (MTX), ATF (ATX), and fuel
    - Runout and contact of pulley and belt
    - Idle speed (See ENGINE TUNE-UP [SKYACTIV-D 1.5].)
    - Engine-driven accessories operation

Note
- If the engine is overhauled and assembled to the vehicle, perform a road test to verify that there is no abnormal vibration and noise.



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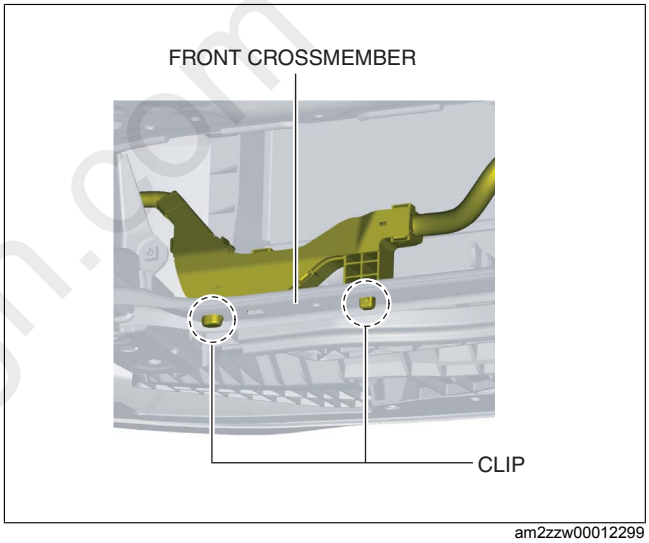
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1	No.1 engine mount rubber, front crossmember component (See No.1 engine mount rubber, front crossmember component removal note.) (See Engine mount installation note.)
2	No.1 engine mount bracket (See Engine mount installation note.)
3	No.4 engine mount bracket (See No.3 engine mount, No.4 engine mount bracket removal note.) (See Engine mount installation note.)
4	No.3 engine mount (See No.3 engine mount, No.4 engine mount bracket removal note.) (See Engine mount installation note.)
5	Engine, transaxle

No.1 engine mount rubber, front crossmember component removal note

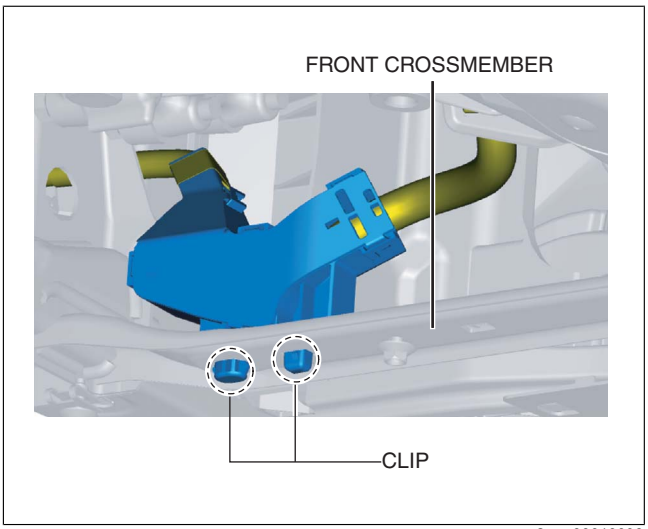
1. Disconnect the service plug. (with i-ELOOP) (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)
2. Disconnect the engine wiring harness from the capacitor. (with i-ELOOP) (See CAPACITOR (i-ELOOP) REMOVAL/INSTALLATION [i-ELOOP].)
3. Remove the clips shown in the figure and set the wiring harness aside. (with i-ELOOP)

MTX



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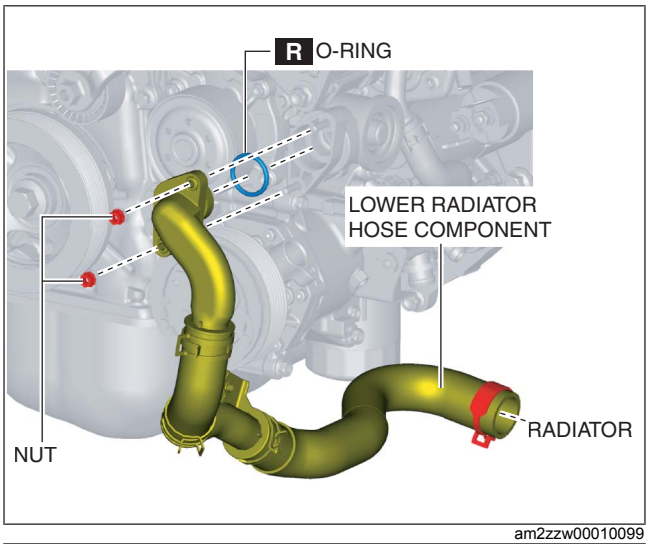
ATX



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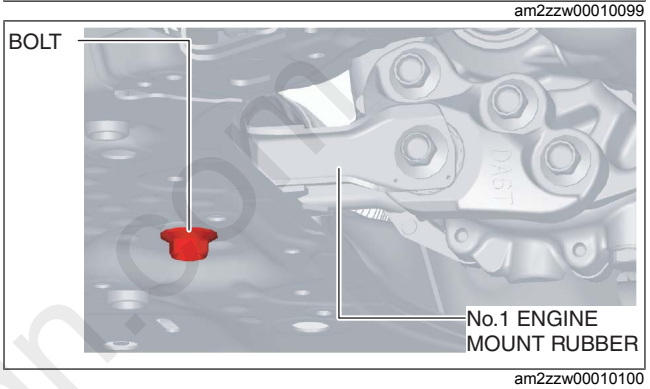
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4. Remove the lower radiator hose component.



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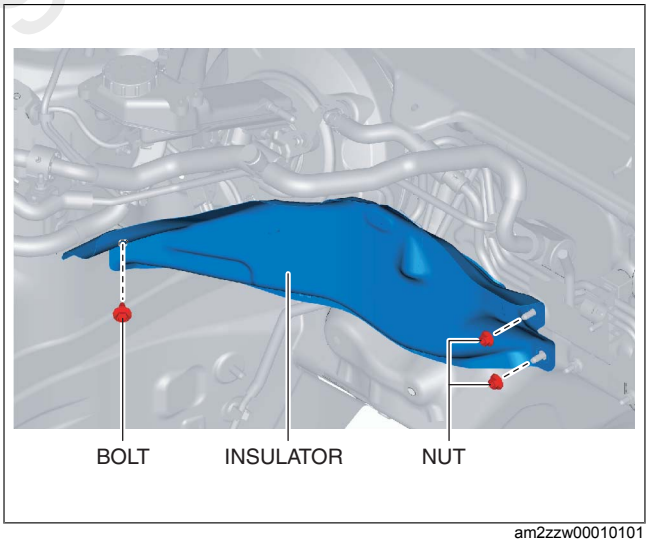
5. Loosen the No.1 engine mount rubber installation bolt (front crossmember side) shown in the figure.
6. Remove the No.1 engine mount rubber and the front crossmember component as a single unit. (See FRONT CROSSMEMBER REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)



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No.3 engine mount, No.4 engine mount bracket removal note

1. Remove the insulator.

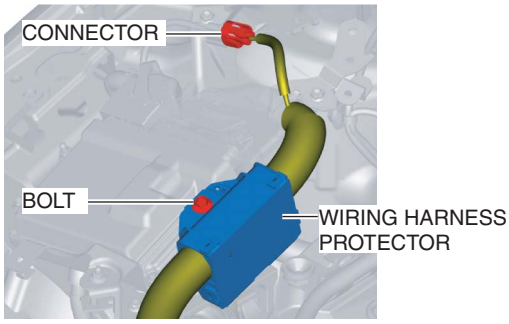


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2. Set the emission wiring harness and PCM component aside using the following procedure:
   (1) Disconnect the PCM connector (140 terminal) on the emission wiring harness side from the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

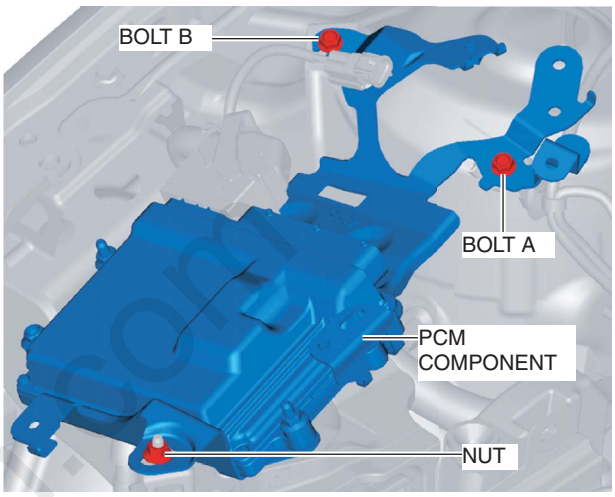
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(2) Remove the bolt and connector shown in the figure and set the emission wiring harness on the engine.



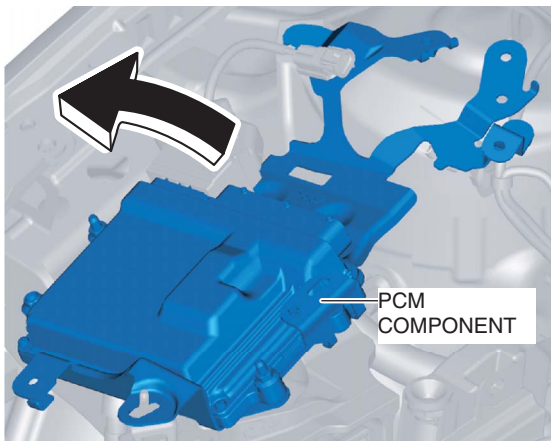
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(3) Remove the bolts and the nut shown in the figure.



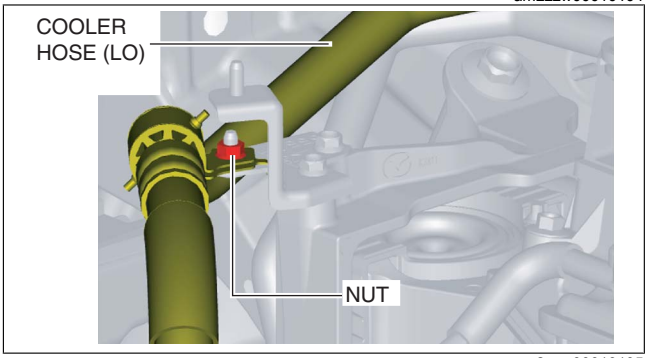
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(4) Move the PCM component in the direction of the arrow shown in the figure with the PCM connector (60 terminal) on the front wiring harness side connected, and set the PCM component aside.



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3. Remove the nut shown in the figure and set the cooler hose (LO) aside.

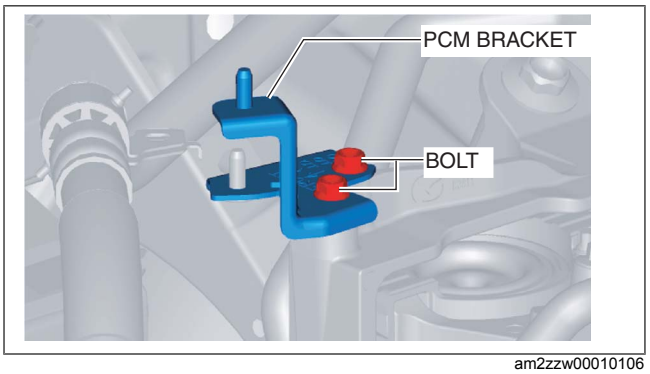


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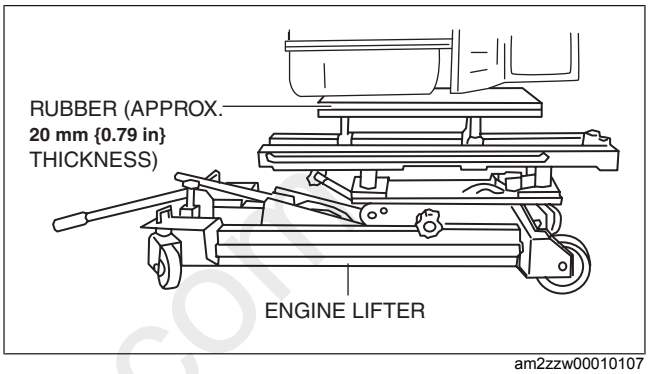
4. Remove the bolts shown in the figure and remove the PCM bracket.
5. Remove the seal plate installed to the underside of the oil pan. (See OIL PAN REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)

Caution
- **When supporting the engine and transaxle, insert rubber of appropriate size (approx. 20 mm {0.79 in} thickness) between the engine lifter and the oil pan, transaxle to prevent deformation of the oil pan and damage of the transaxle.**



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6. Secure the engine and transaxle using a commercially available engine lifter.
7. Remove the No.4 engine mount bracket.
8. Remove the No.3 engine mount.



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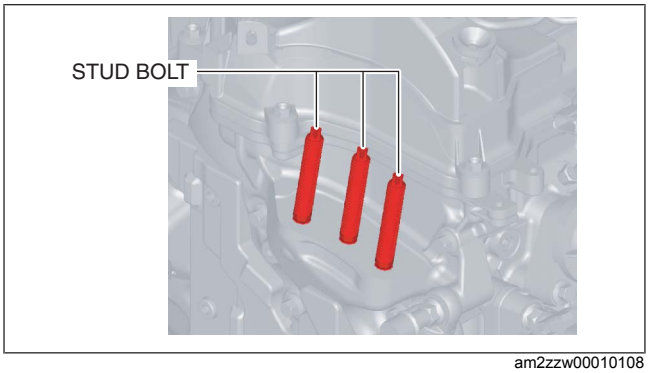
Engine mount installation note

Caution
- **Slots have been adopted for the No.3 engine mount and No.4 engine mount bracket installation holes. Tighten the installation bolts and nuts so that they are centered on the slots. Otherwise, engine noise or vibration could increase.**

Note
- When loosening the nuts of the No.3 engine mount and the No.4 engine mount bracket, the engine front cover and the transaxle stud bolt may loosen.

1. Tighten the engine front cover stud bolts.

Tightening torque
7—13 N·m {72—132 kgf·cm, 62—115 in·lbf}



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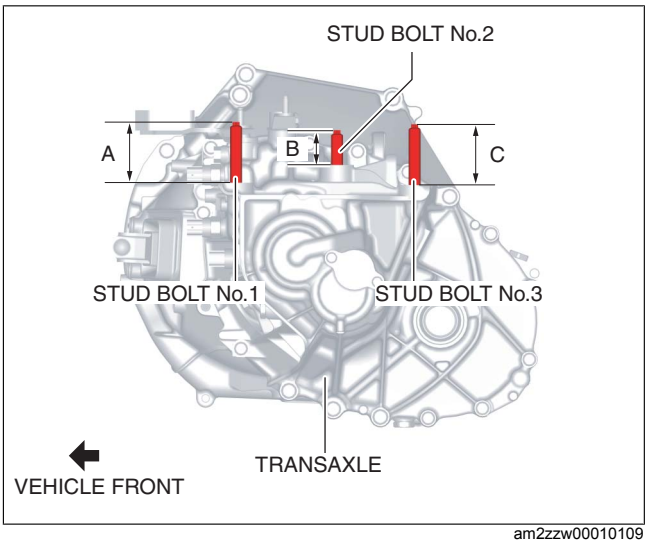
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2. Measure the projection of the transaxle stud bolts.

Projection A of stud bolt No.1
70.7—72.7 mm {2.79—2.86 in}
Projection B of stud bolt No.2
40.0—42.0 mm {1.58—1.65 in}
Projection C of stud bolt No.3
70.7—72.7 mm {2.79—2.86 in}

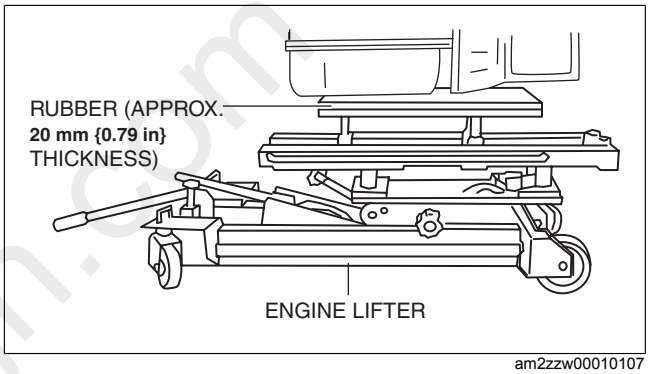
- If the projection amount is not within the range, adjust the projection amount of the stud bolt.

Caution
- **When supporting the engine and transaxle, insert rubber of appropriate size (approx. 20 mm {0.79 in} thickness) between the engine lifter and the oil pan, transaxle to prevent deformation of the oil pan and damage of the transaxle.**



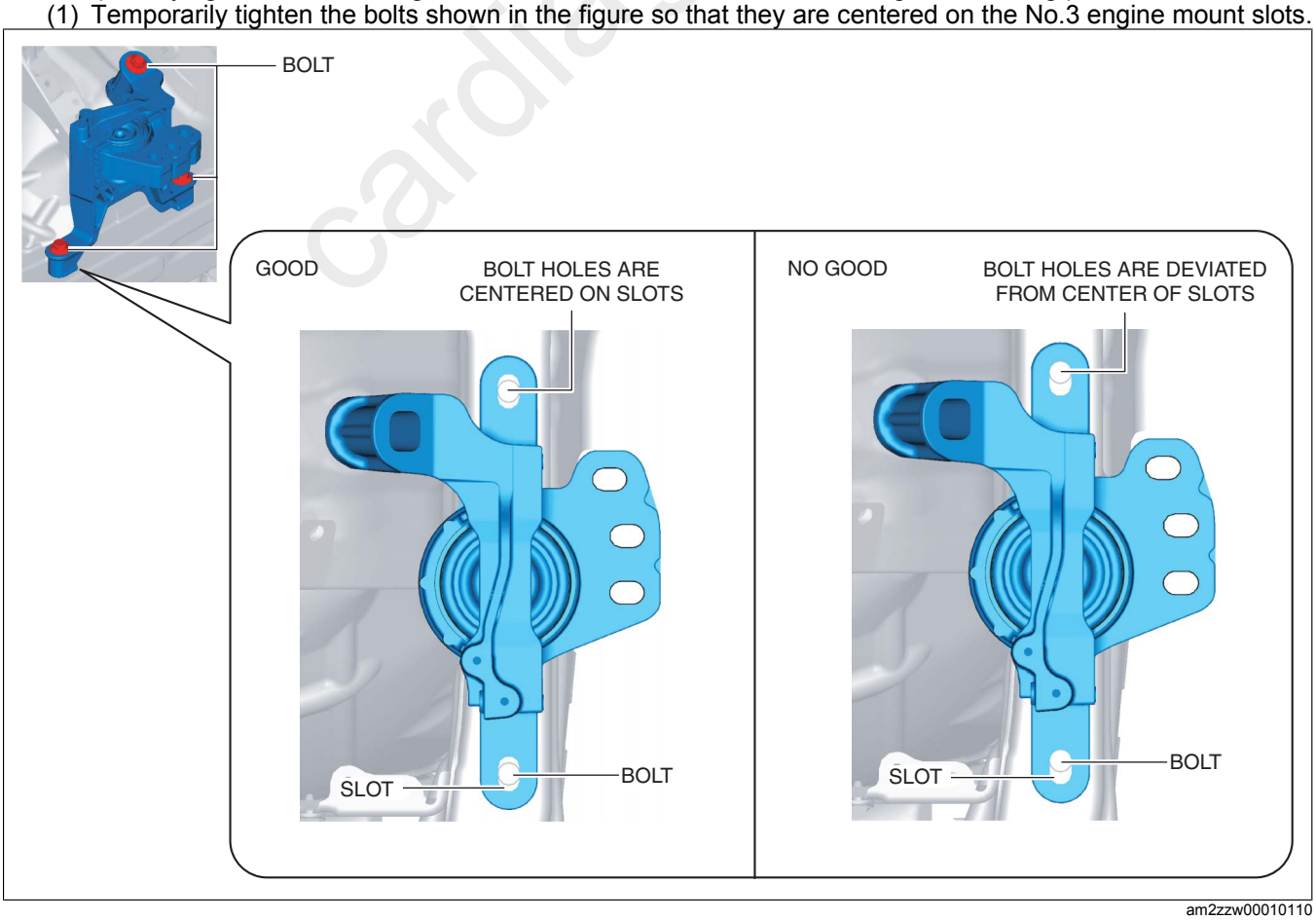
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3. Secure the engine and transaxle using a commercially available engine lifter.



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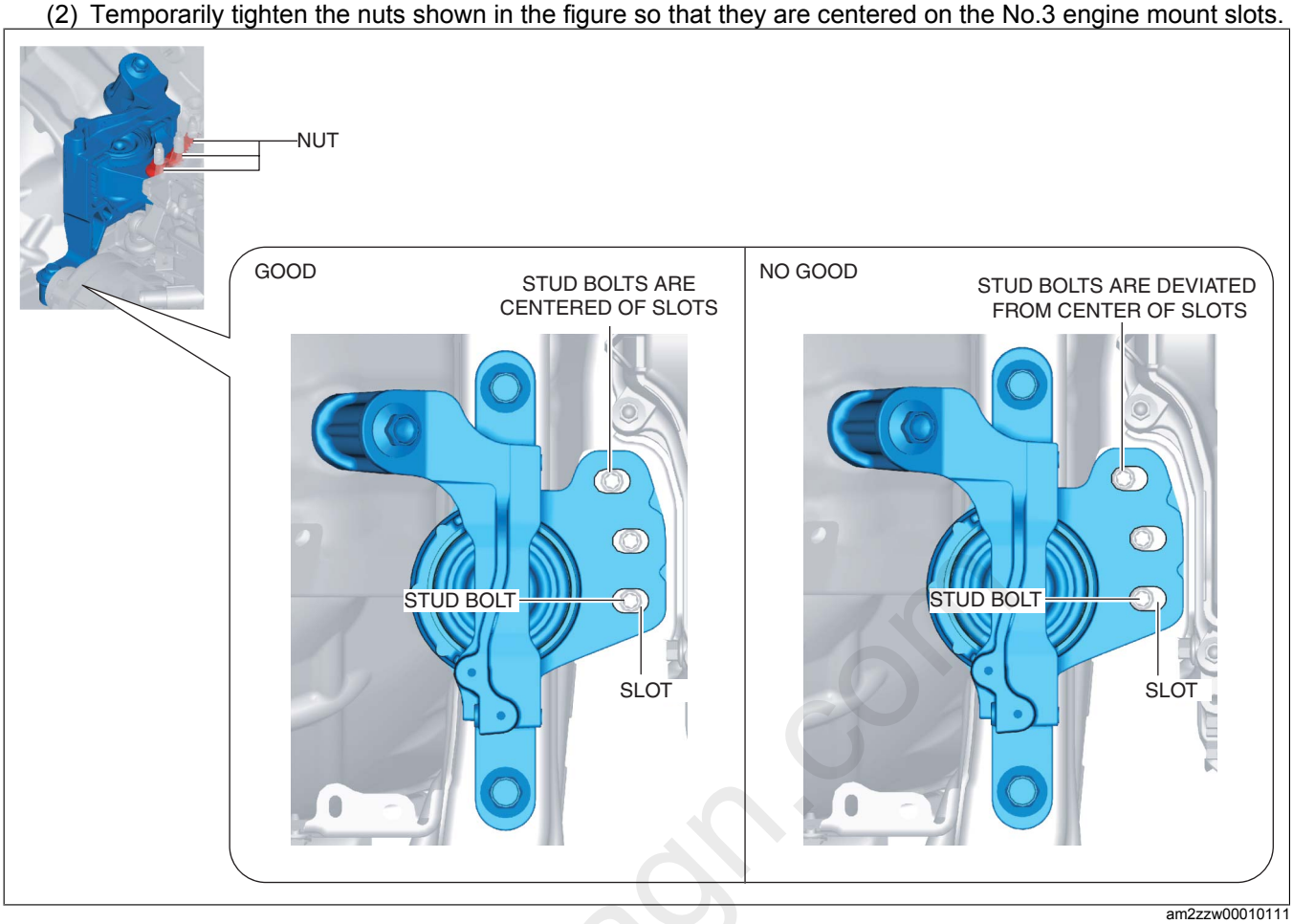
4. Temporarily tighten the No.3 engine mount installation bolts and nuts using the following procedure:
(1) Temporarily tighten the bolts shown in the figure so that they are centered on the No.3 engine mount slots.



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(2) Temporarily tighten the nuts shown in the figure so that they are centered on the No.3 engine mount slots.

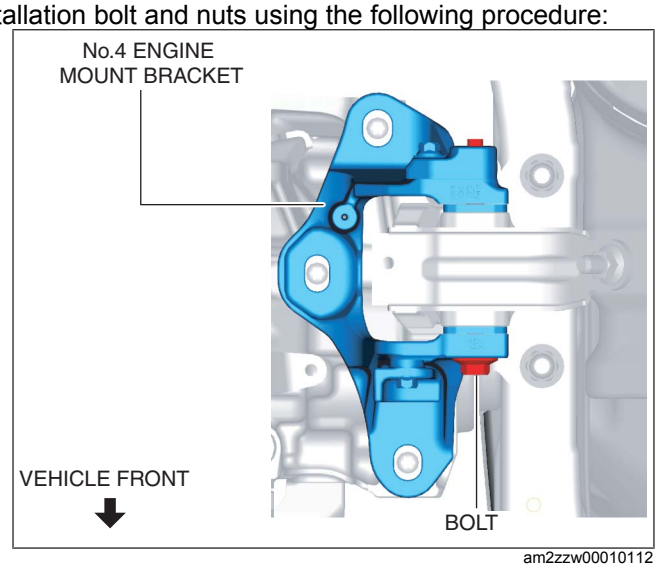


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Note

- If the stud bolts are deviated from the center of the slots, align the stud bolts with the center of the slots while slightly moving the engine, and temporarily tighten the installation nuts.

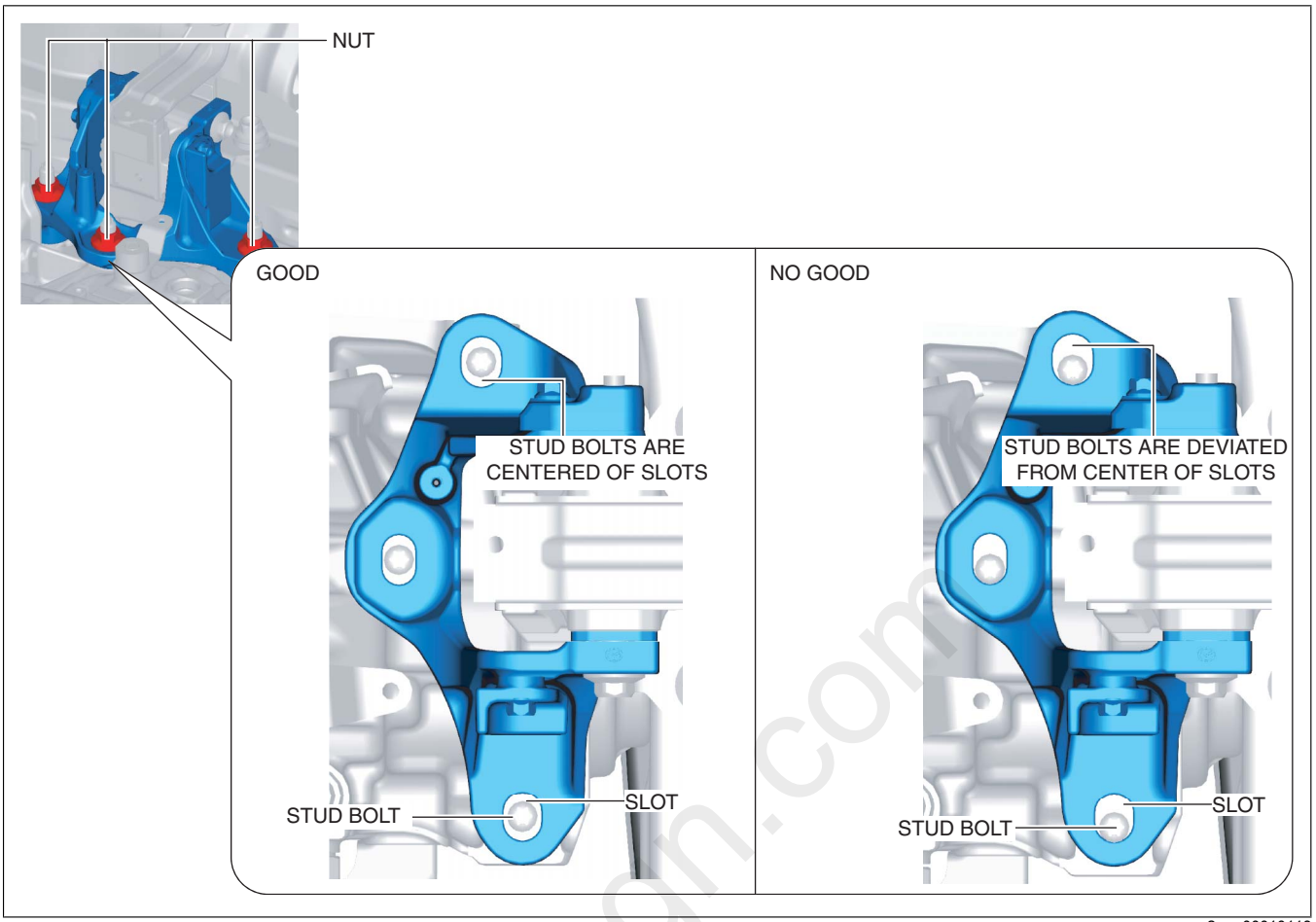
5. Temporarily tighten the No.4 engine mount bracket installation bolt and nuts using the following procedure:
   (1) Temporarily tighten the bolt shown in the figure.



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   (2) Temporarily tighten the nuts shown in the figure so that they are centered on the No.4 engine mount bracket slots.

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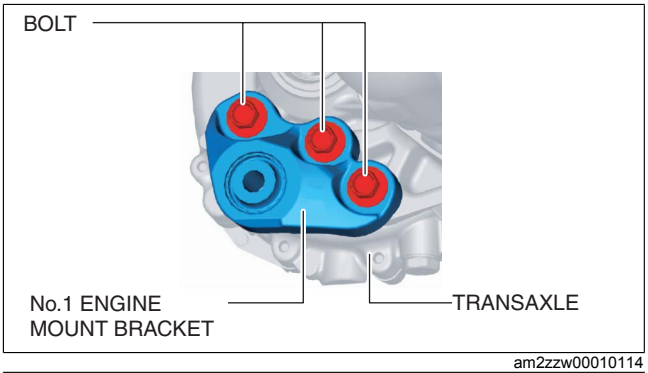


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Note

- If the stud bolts are deviated from the center of the slots, align the stud bolts with the center of the slots while slightly moving the transaxle, and temporarily tighten the installation nuts.

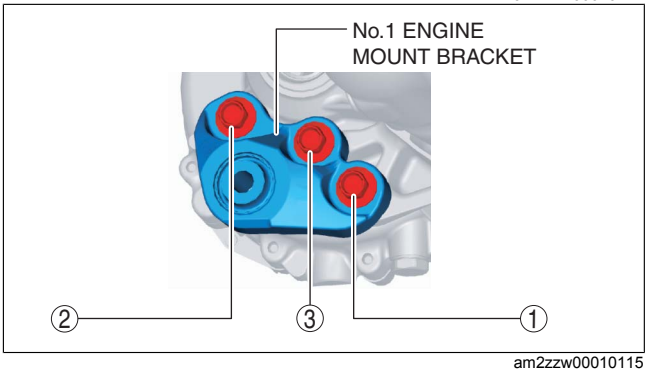
6. Install the No.1 engine mount bracket and temporarily tighten the bolts shown in the figure.



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7. Tighten the No.1 engine mount bracket installation bolts in the order shown in the figure.

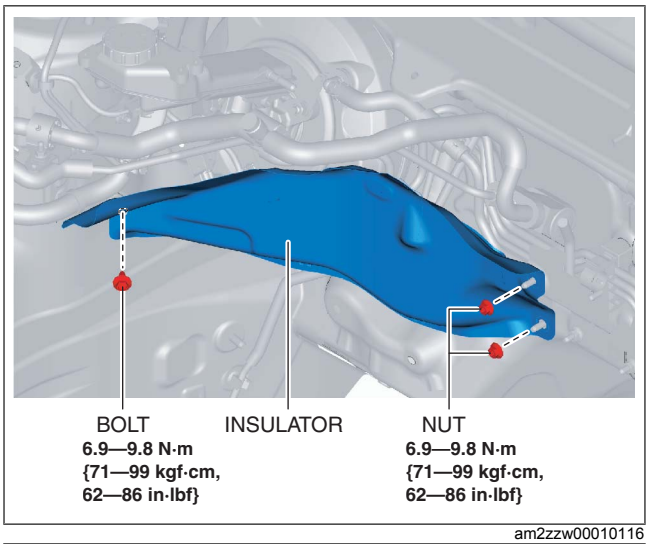
Tightening torque
84—98 N·m {8.6—9.9 kgf·m, 62—72 ft·lbf}



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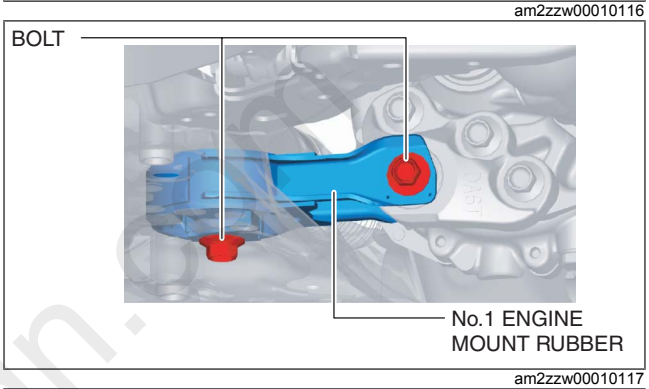
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8. Install the insulator
9. Install the No.1 engine mount rubber and the front crossmember component as a single unit. (See FRONT CROSSMEMBER REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)



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10. Temporarily tighten the No.1 engine mount rubber installation bolts.

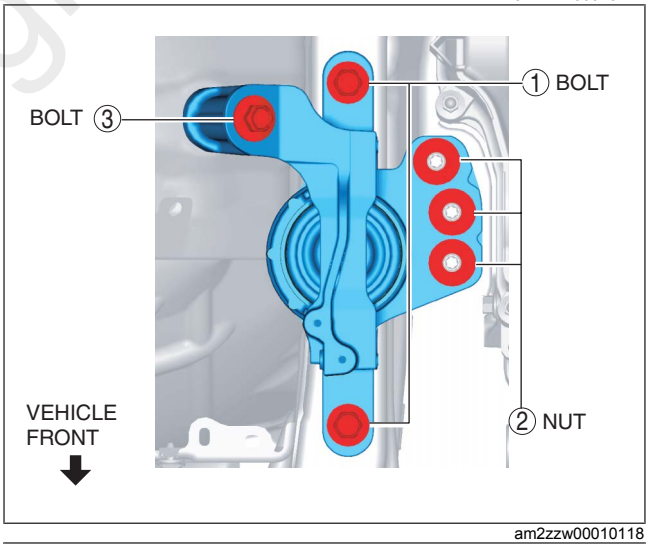


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11. Tighten the No.3 engine mount installation bolts and nuts in the order shown in the figure.

Tightening torque

Installation position	Tightening torque
1	76—95 N·m {7.8—9.6 kgf·m, 57—70 ft·lbf}
2	82—95 N·m {8.4—9.6 kgf·m, 61—70 ft·lbf}
3	49—65 N·m {5.0—6.6 kgf·m, 37—47 ft·lbf}

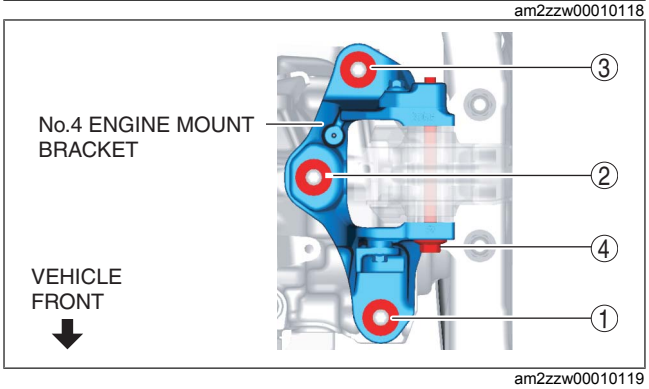


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12. Tighten the No.4 engine mount bracket installation bolt and nuts in the order shown in the figure.

Tightening torque

Installation position	Tightening torque
1, 2, 3	92—116 N·m {9.4—11 kgf·m, 68—85 ft·lbf}
4	81—99 N·m {8.3—10 kgf·m, 60—73 ft·lbf}



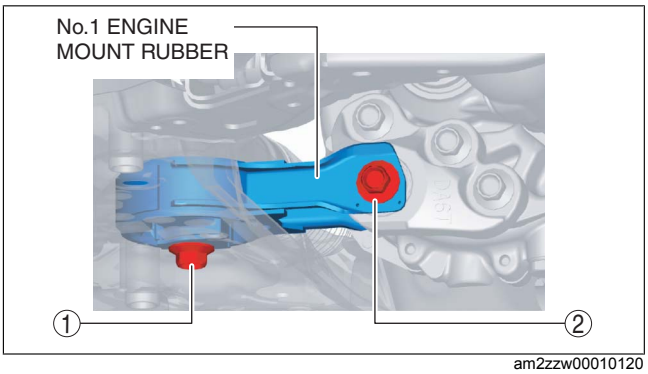
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13. Tighten the No.1 engine mount rubber installation bolts in the order shown in the figure.

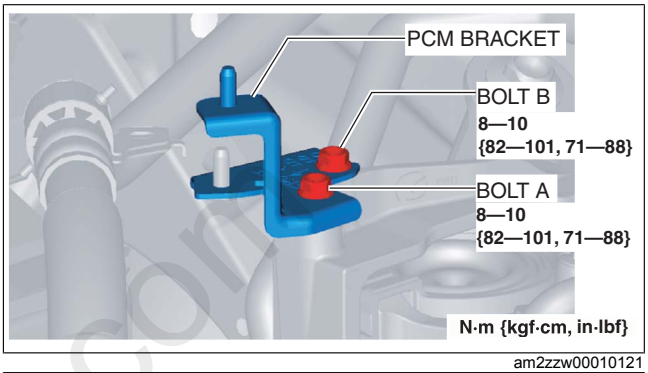
Tightening torque

Installation position	Tightening torque
1	141—172 N·m {15—17 kgf·m, 104—126 ft·lbf}
2	130—164 N·m {14—16 kgf·m, 96—120 ft·lbf}



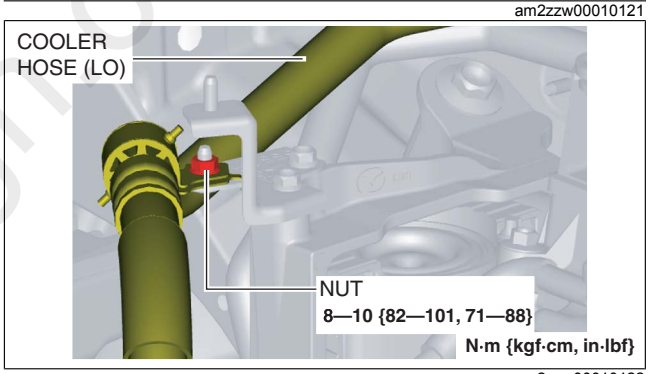
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14. Install the seal plate. (See OIL PAN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
15. Install the PCM bracket using the following procedure:
    (1) Temporarily tighten the bolt A.
    (2) Tighten the bolts in the order of B and A.



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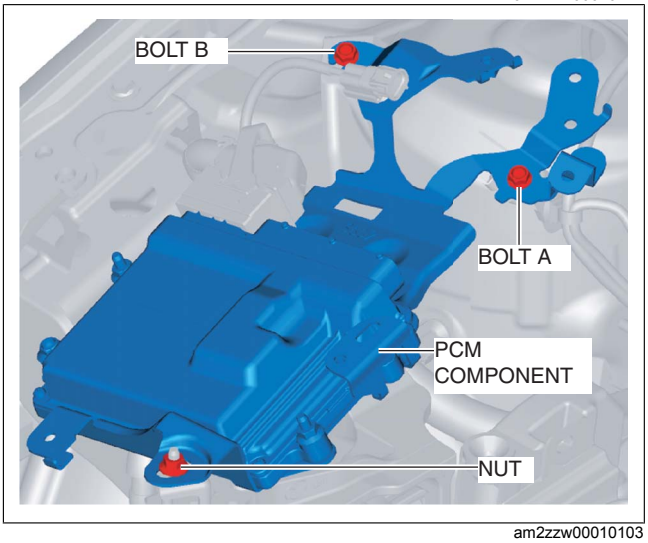
16. Install the nut shown in the figure and install the cooler hose (LO).



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17. Install the PCM component using the following procedure:
    (1) Return the PCM component that was set aside to its original position.
    (2) Temporarily tighten the bolt A.
    (3) Tighten the PCM component installation bolts and the nut in the order of bolt B, the nut, and bolt A.

    Tightening torque
    Bolt: 9—10 N·m {92—101 kgf·cm, 80—88 in·lbf}
    Nut: 8—10 N·m {82—101 kgf·cm, 71—88 in·lbf}

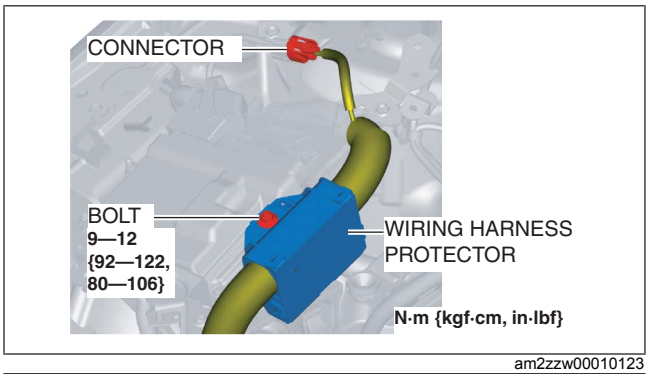


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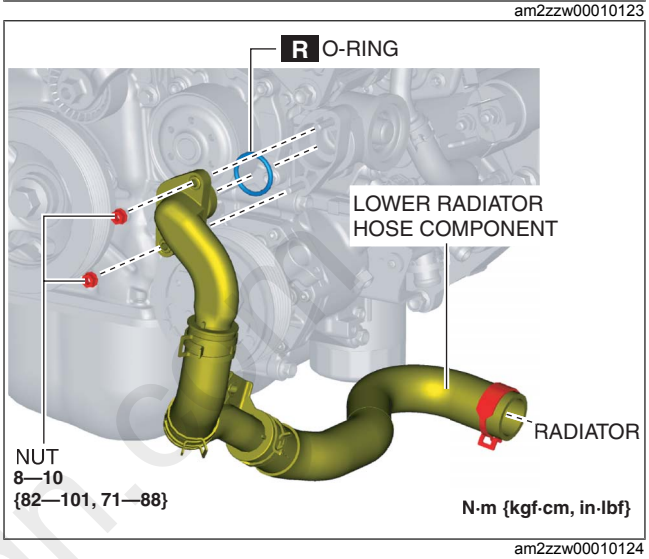
(4) Install the bolt and connector as shown in the figure.
(5) Connect the PCM connector (140 terminal) on the emission wiring harness side to the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

Caution
- **Do not apply oil (such as engine oil, ATF) to the O-ring. Otherwise, the O-ring could swell causing a seal malfunction.**



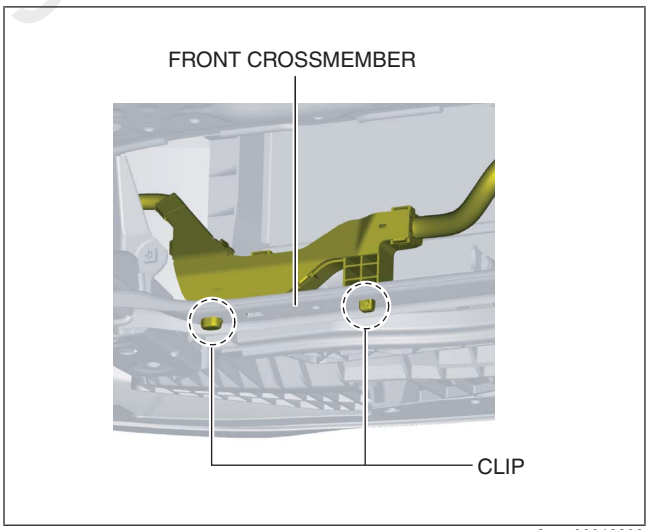
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18. Install the lower radiator hose component.



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19. Install the wiring harness clips shown in the figure to the front crossmember. (with i-ELOOP)
MTX

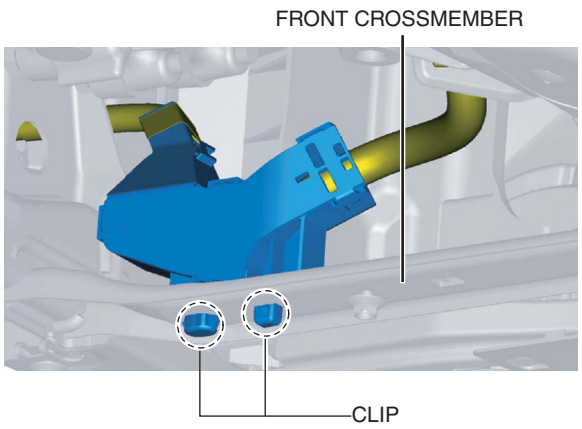


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ATX

20. Connect the wiring harness to the capacitor. (with i-ELOOP) (See CAPACITOR (i-ELOOP) REMOVAL/ INSTALLATION [i-ELOOP].)
21. Install the service plug. (with i-ELOOP) (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].)



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